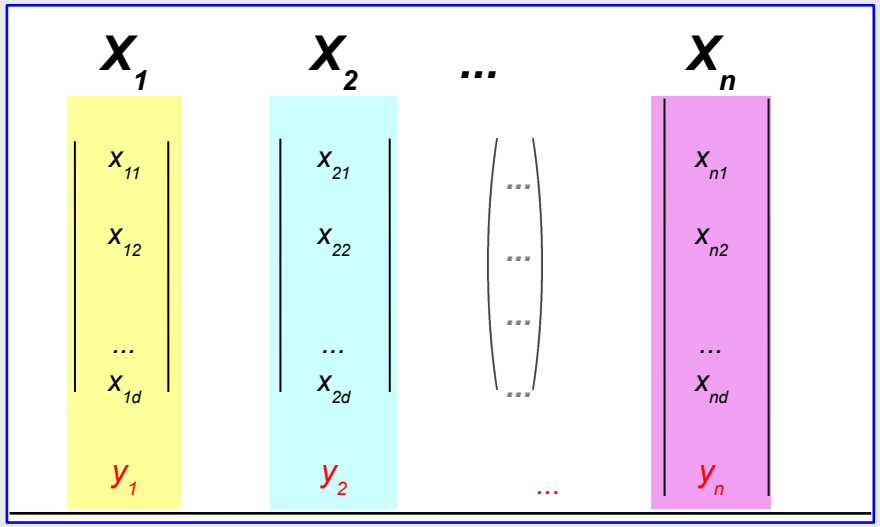
Matrix Notation for Linear Regression and Classification (N- Samples)

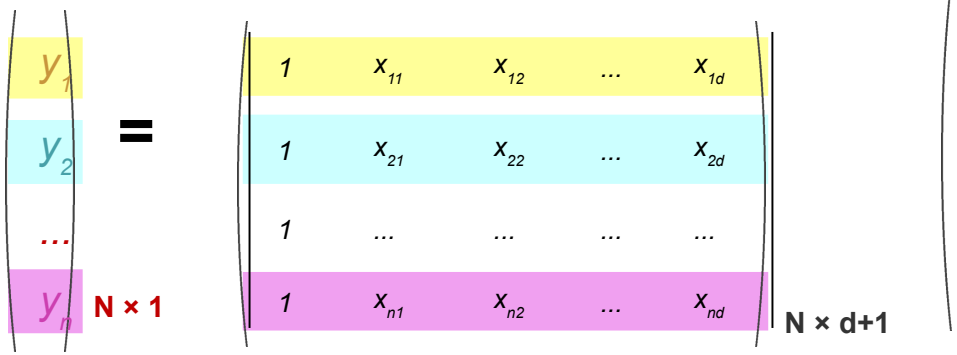
N Training Samples



Remember

$$\underline{y} = \underline{W}^T\underline{X} = \underline{X}^T\underline{W}$$

In General: **N >>> d**



$$\underline{Y} = \underline{X}\,\underline{W}$$

Where: $\underline{X}_{N\times d+1}$ **is a matrix**

$\underline{Y}_{N\times 1}$ **is labels vector**